MATH317 · NUMERICAL METHODS

Chapter 1.2 Taylor Series

Midterm summary · Lecture 3

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01 What is a Taylor series?

Functions such as e^x and $\sin x$ are hard to compute directly. A **Taylor series** rewrites the function as a sum of simple terms, each one a power and a division.

$$f(x) = f(c) + f'(c)(x - c) + \frac{f''(c)}{2!}(x - c)^2 + \frac{f'''(c)}{3!}(x - c)^3 + \dots$$

Symbol	Meaning
c	The point we expand around (the **Expansion point**).
$f^{(k)}(c)$	The k-th derivative evaluated at c.
$k!$	k factorial. Example: $3! = 6$.

Maclaurin series

When $c = 0$ the series is called a **Maclaurin series**.

02 The familiar series

These five series should be known:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots \qquad (|x| < 1)$$

$$\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \dots \qquad (0 < x \le 2)$$

The first four are expanded about `c = 0`. The last one is about `c = 1`.

Example 1: computing ln(1.1)

Put `x = 1.1` in the series for `ln x`, so `x − 1 = 0.1`. Take five terms:

$$0.1 - \frac{0.1^2}{2} + \frac{0.1^3}{3} - \frac{0.1^4}{4} + \frac{0.1^5}{5}$$

$$= 0.1 - 0.005 + 0.000333 - 0.000025 + 0.000002$$

$$= 0.0953103$$

The true value is `0.0953102`. Five terms were enough because `1.1` is close to `c = 1`.

Example 2: computing e^8

Put `x = 8` in the series for `eˣ` and take six terms:

$$1 + 8 + \frac{64}{2} + \frac{512}{6} + \frac{4096}{24} + \frac{32768}{120} = 570.07$$

True value: $e^8 = 2980.96$

The general rule

A Taylor series converges rapidly near the expansion point and slowly (or not at all) far from it.

A True/False statement about this rule appeared in midterm 472.

03 Taylor series of a polynomial

When the function is a polynomial the series stops after finitely many terms, and the result is exact.

Method 1: derivatives

Example: $f(x) = x^3 - 2x^2 + 4x - 1$ about $c = 2$.

$$f(x) = x^3 - 2x^2 + 4x - 1 \qquad f(2) = 8 - 8 + 8 - 1 = 7$$

$$f'(x) = 3x^2 - 4x + 4 \qquad f'(2) = 12 - 8 + 4 = 8$$

$$f''(x) = 6x - 4 \qquad f''(2) = 8 \qquad \to \frac{8}{2!} = 4$$

$$f'''(x) = 6 \qquad f'''(2) = 6 \qquad \to \frac{6}{3!} = 1$$

$$\mathbf{f(x) = 7 + 8(x-2) + 4(x-2)^2 + (x-2)^3}$$

Method 2: complete Horner (faster)

Apply Horner repeatedly with the same r. Each time, the last number of the row is the next coefficient, and we continue with the rest of the row.

Slide example: $p(x) = x^4 - 4x^3 + 7x^2 - 5x + 2$ about $r = 3$.

	1	−4	7	−5	2	
3 |		3	−3	12	21	
	1	−1	4	7 |	23	→ c0 = 23
		3	6	30		
	1	2	10 |	37		→ c1 = 37
		3	15			
	1	5 |	25			→ c2 = 25
		3				
	1 |	8				→ c3 = 8
	1					→ c4 = 1

$$p(x) = (x-3)^4 + 8(x-3)^3 + 25(x-3)^2 + 37(x-3) + 23$$

How to check

c_0 must equal $p(r)$. Here $p(3) = 81 - 108 + 63 - 15 + 2 = 23$.

Appeared in the midterm

This question appeared in three midterms (462, 471, 472) for five marks. In 462 and 471 the question fixed the method in words: "Using the complete Horner's algorithm", so you must solve it with Horner. In 472 it named no method ("What is the Taylor series..."), so you may choose. Learn Horner because it is usually the one required, and use derivatives to check.

04 Taylor's theorem and the remainder

In practice we stop the series after some terms (**Truncation**). The difference between the function and the truncated series is the **Remainder** or **Error term**.

$$f(x) = \Sigma\ (k = 0 \to n)\ \ \frac{f^{(k)}(c)}{k!}\ (x - c)^k\ \ +\ \ E_{n+1}$$

$$E_{n+1} = \frac{f^{(n+1)}(\xi)}{(n+1)!}\ \ \cdot\ \ (x - c)^{n+1}$$

ξ is an unknown point between c and x.

The second form, in terms of h

Replace x by $x + h$ and c by x:

$$f(x + h) = f(x) + h\ f'(x) + \frac{h^2}{2!}\ f''(x) + \ldots + E_{n+1}$$

$$E_{n+1} = O(h^{n+1})$$

$O(h^{n+1})$ means the error is at most a constant times h^{n+1} as h becomes small.

When does the series represent the function?

When the remainder goes to zero as n grows.

Function	Result
e^x	The series represents the function for every x.
$\ln(1 + x)$	It represents it for $-1 < x \leq 1$ (the slide proves the case $0 \leq x \leq 1$). For $x > 1$ it does not converge.

05 Mean-Value Theorem

It is the special case `n = 0` of Taylor's theorem.

$$f(b) - f(a) = (b - a)\, f'(\xi) \qquad \xi \text{ in } (a, b)$$

In words: somewhere between `a` and `b` the slope of the tangent equals the slope of the line joining the two ends.

Appeared in the midterm

A statement in midterm 472 said the Mean-Value Theorem is a special case of Taylor's theorem "with zero derivatives". The answer was False: it is the case `n = 0`, not "zero derivatives".

06 Alternating series

A series whose terms alternate in sign: plus, minus, plus...

The theorem

If the terms decrease to zero, the series converges. If we stop at some term, the error is at most the first term we left out.

$$|\, S - S_n \,| \le a_{n+1}$$

Slide example: how many terms for sin 1?

The error must be below $\frac{1}{2} \times 10^{-6}$.

$$\sin 1 = 1 - \frac{1}{3!} + \frac{1}{5!} - \frac{1}{7!} + \ldots$$

Stop after n terms → error ≤ $\frac{1}{(2n+1)!}$

Need: $\frac{1}{(2n+1)!} < \frac{1}{2} \times 10^{-6}$ → $(2n+1)! > 2 \times 10^{6}$

9! = 362,880 (too small)

11! = 39,916,800 (enough) → 2n + 1 = 11

n = 5 terms

07 Common mistakes

Mistake	Fix
Forgetting to divide by `k!`	The coefficient is $\frac{f^{(k)}(c)}{k!}$, not the derivative alone.
Writing `(x + 3)` instead of `(x − 3)`	The form is `(x − c)`. If `c = −3` it becomes `(x + 3)`.
Saying the series converges fast far from the point	It converges fast near the point.
Reversing the coefficients in complete Horner	The first number produced is c_0 (the constant term).

08 Check yourself

1. Find the Taylor expansion of $P(x) = x^3 - 2x^2 + 4x - 3$ about $r = 2$ using complete Horner.
2. Find the Taylor expansion of $P(x) = 2x^3 - 3x^2 + 4$ about $r = -1$.
3. How many terms of the series for $\ln(1 + x)$ are needed to compute $\ln 1.1$ with error below $\frac{1}{2} \times 10^{-8}$?

Solutions

```
1)        1     −2      4     −3
  2 |             2      0      8
          1      0      4  |   5        c0 = 5
                 2      4
          1      2  |   8              c1 = 8
                 2
          1  |   4                     c2 = 4
          1                           c3 = 1
```

$$P(x) = (x-2)^3 + 4(x-2)^2 + 8(x-2) + 5$$

```
2)        2     −3      0      4
 −1 |            −2      5     −5
          2     −5      5  |  −1        c0 = −1
                −2      7
          2     −7  |  12              c1 = 12
                −2
          2  |  −9                     c2 = −9
          2                           c3 = 2
```

$$P(x) = 2(x+1)^3 - 9(x+1)^2 + 12(x+1) - 1$$

3) $\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots$ with $x = 0.1$

After n terms the error $\leq \frac{(0.1)^{n+1}}{n+1}$

n = 6: $\frac{10^{-7}}{7} = 1.43 \times 10^{-8}$ (not enough)

n = 7: $\frac{10^{-8}}{8} = 1.25 \times 10^{-9}$ (less than 5×10^{-9})

7 terms